6.3 Logarithms and Logarithmic Functions

Name _____ Period ____ Date _____

Algebra II 6.3 · Student sheet

I can…

- rewrite logarithmic and exponential equations.
- evaluate logarithms, with and without a calculator.
- find inverses and graph logarithmic functions.

Vocabulary · fill in the missing word

Logarithm — $\log_b y = x$ _____ $b^x = y$ ($b > 0$, $b \neq 1$): the exponent you put on b to get y

Common logarithm — $\log x$ = _____

Natural logarithm — $\ln x$ = _____

Logarithmic function — $f(x) = \log_b x$, the inverse of $g(x) = b^x$; domain $x > 0$, _____ $x = 0$

Part 1 · Exponent hunt

Find each missing exponent. Then write the equation as a logarithm: "log base __ of __ = __."

Equation	Exponent	As a logarithm
2^? = 8		
3^? = 81		
10^? = 1000		
5^? = 1/25		
4^? = 2		
7^? = 1		

Can any exponent on 2 give 0 or a negative number? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Definition of a logarithm: $\log_b y = x$ means _____. A logarithm is an _____.

Example 1

a. Rewrite $\log_3 81 = 4$ in exponential form. b. Rewrite $6^{-1} = \frac{1}{6}$ in logarithmic form.

Special logarithms: $\log_b 1$ = _____ $\log_b b$ = _____ log x means log base _____ ln x means log base _____

Example 2

Evaluate each logarithm.

a. $\log_4 64$ b. $\log_5 0.2$ c. $\log_{(\frac{1}{3})} 9$ d. $\log_{36} 6$

Example 3

Evaluate with a calculator. Round to three decimal places.

a. log 8 b. ln 0.3

Inverse properties: Inverse properties: $\log_b b^x$ = _____ and b^($\log_b$ x) = _____

Example 4

Simplify each expression.

a. 10^(log 4) b. $\log_5 25^x$ c. e^(ln 7)

Logarithmic functions: $f(x) = \log_b x$ is the inverse of _____. Its domain is _____ and its asymptote is _____.

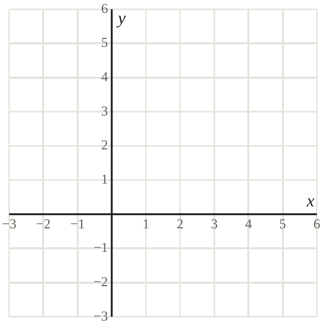
Example 5

Find the inverse of each function.

a. $y = 6^x$ b. $y = \ln(x + 3)$

Example 6

Graph $f(x) = \log_2 x$ using its inverse, $g(x) = 2^x$.



Part 3 · Partner practice

Partner 1

Evaluate $\log_2 32$ and $\log_7 \left(\frac{1}{49}\right)$.

Partner 2

Find the inverse of $y = 4^x + 1$.